VortexTech AI/ML Internship — Progress Report (Weeks 1–2)

Track: AI & ML Internship **Period covered:** Week 1 (Data Cleaning & EDA) and Week 2 (Classification Model) **Dataset used:** Titanic passenger dataset (891 rows), sourced from the public seaborn-data repository

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Week 1: Data Cleaning and Exploratory Analysis

Objective: Take a raw, messy dataset and prepare it for analysis using Pandas.

What was done: - Loaded the Titanic dataset and inspected its structure (`.info()`, `.describe()`) - Identified data quality issues: `age` missing in ~20% of rows, `deck` missing in ~77%, `embarked`/`embark_town` missing 2 rows each, and 107 rows flagged as exact duplicates - Cleaned the dataset with justified, column-specific decisions: - `age` → filled with median (robust to outliers) - `embarked` / `embark_town` → filled with mode (negligible impact, only 2 rows) - `deck` → recoded missing values as `"Unknown"` rather than dropped, since ~77% missingness made imputation unreliable - Flagged duplicate rows were reviewed rather than auto-dropped, since the dataset has no unique passenger ID - Produced summary statistics: numeric `.describe()`, class value counts, and survival rate by sex/class - Produced two visualizations: a histogram of passenger age and a bar chart of survival rate by class

Key finding: Survival was strongly associated with passenger class and sex — consistent with the historical record of the disaster.

Deliverable: `vortextech-aiml-week1` — notebook, dataset, and README on GitHub.

Week 2: Build a Classification Model

Objective: Train and evaluate a binary classification model using scikit-learn.

What was done: - Reused the cleaned Week 1 dataset, with `survived` (0/1) as the target - Selected features: passenger class, sex, age, family size, fare, and port of embarkation - One-hot encoded categorical features (`sex`, `embarked`) using `pd.get_dummies()` - Split the data 80/20 into training and test sets - Trained a **Logistic Regression** model and evaluated it with accuracy, precision, recall, and F1-score - Trained a **Decision Tree** as a comparison model

Results:

Model	Accuracy	F1-score
Logistic Regression	~0.80	~0.75
Decision Tree	~0.75	~0.69

Key finding: Logistic Regression outperformed the Decision Tree, suggesting the relationship between features and survival is fairly linear/simple rather than requiring complex branching rules. The model isn't perfect — it likely struggles with passengers whose class/sex/age combination doesn't clearly point toward survival or not.

Ideas for improvement: engineer new features (e.g. family size from `sibsp` + `parch`, or title extracted from name), tune hyperparameters, or try an ensemble method like Random Forest.

Deliverable: `vortextech-aiml-week2` — notebook and README on GitHub.

Summary

Across the first two weeks, the work progressed from raw data handling to a working, evaluated machine learning model:

1. **Week 1** built the foundation — a clean, well-understood dataset with documented cleaning decisions.
2. **Week 2** used that foundation to train and compare two classification models, achieving ~80% accuracy with Logistic Regression.

Both weeks' notebooks are fully reproducible, documented with markdown explanations at each step, and pushed to public GitHub repositories as required.